A Lasserre SDP Lower Bound for the Autoconvolution Constant C_{1a}

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Abstract

We prove that the autoconvolution constant

$$C_{1a} \;=\; \inf_{f} \frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2},$$

taken over nonnegative functions supported on $(-\frac{1}{4}, \frac{1}{4})$, satisfies $C_{1a} \geq 1.3$. The proof reduces the infinite-dimensional optimization to the polynomial program

$$\mathrm{val}(d) \;=\; \min_{\mu \in \Delta_d} \; \max_{W} \, \mu^\top M_W \, \mu,$$

where Δ_d is the standard simplex and $\{M_W\}$ is a finite family of nonnegative matrices encoding windowed autoconvolution test values. We show $\mathrm{val}(d) \leq C_{1a}$ for every d, then certify $\mathrm{val}(d) \geq 1.3$ at $d = 16$ via the order-3 Lasserre hierarchy [Las01]. At higher resolutions we replace the full moment cone by its correlative-sparsity restriction [WKKM06], which preserves soundness while accommodating numerical corroboration at $d \in \{32, 64, 128\}$. The external dependency of the proof is a single finite SDP at $(d, k) = (16, 3)$ whose dual solution we record as the certificate.

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1. Introduction

1.1. The Constant C_{1a}

All source code and the SDP certificate are available at the accompanying repository. Let

$$\mathcal{F} \;=\; \left\{ f \in L^1(\mathbb{R}) : f \geq 0,\ \mathrm{supp}(f) \subseteq \left(-\tfrac{1}{4}, \tfrac{1}{4}\right),\ \int_{\mathbb{R}} f > 0 \right\}.$$

For $f \in \mathcal{F}$ the *autoconvolution* $(f * f)(t) = \int f(x) f(t-x)\, dx$ is continuous, nonnegative, and supported on $[-\frac{1}{2}, \frac{1}{2}]$. The *autoconvolution ratio* and *autoconvolution constant* are

$$R(f) \;=\; \frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2}, \qquad C_{1a} \;=\; \inf_{f \in \mathcal{F}} R(f).$$

The constant C_{1a} is connected to the asymptotic density of Sidon and generalized Sidon sets in additive combinatorics [ET41, MO06, CRV10, MO09, CS17], and its numerical value has resisted substantial improvement for nearly a decade.

Bound	Value	Reference
Published lower bound	≥ 1.2802	Cloninger & Steinerberger [CS17]
SDP lower bound	≥ 1.3	This work
Cascade lower bound	≥ 1.4	Bajaj [Baj26]
Published upper bound	≤ 1.50992	Matolcsi & Vinuesa [MV10]

Table 1: Published bounds on C_{1a} in the normalization used here, together with the present result.

1.2. Known Bounds and the Present Result

The branch-and-prune framework of Cloninger and Steinerberger [CS17] produces lower bounds by exhaustively testing every lattice point in a discretized mass simplex against windowed quadratic inequalities. This framework is sharp in principle, but scales exponentially with the dimension d of the partition. The *cascade* extension of [Baj26] pushed that exponential out to $d = 128$ with $m = 20$ grid points via symmetry reduction and dynamic window corrections.

The present paper takes a fundamentally different route. Rather than exhausting the simplex, we observe that the bound of [CS17] reduces to a single polynomial optimization problem on the simplex,

$$\mathrm{val}(d) \;=\; \min_{\mu \in \Delta_d} \; \max_{W \in \mathcal{W}_d} \; \mu^\top M_W \mu,$$

where $\mathcal{W}_d$ enumerates admissible windows and M_W is the associated nonnegative matrix. Any lower bound on $\mathrm{val}(d)$ is a lower bound on C_{1a} (Theorem 2.3). We certify such a lower bound via the Lasserre semidefinite hierarchy, which produces rigorous certificates without enumerating any of the lattice points that the cascade relies on.

Theorem 1.1 (Main result)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be nonnegative with $\mathrm{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4})$ and $\int f > 0$. If $\|f * f\|_{L^\infty} < \infty$, then*

$$R(f) \;=\; \frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2} \;\geq\; 1.3.$$

In particular, $C_{1a} \geq 1.3$.

The proof reduces to a finite SDP certificate at $(d, k) = (16, 3)$, where k is the Lasserre order. A looser variant at $d = 8$, $k = 3$ reproduces the 1.2802 bound of [CS17]; the improvement to 1.3 requires moment matrices of size 969, solved by MOSEK [MOS24] in under ten minutes on a workstation. Auxiliary runs at $d \in \{32, 64, 128\}$ use correlative sparsity [WKKM06] and lie within the same soundness framework; at $d = 16$ the sparse restriction is redundant with the full hierarchy and is not used.

1.3. Proof Strategy at a High Level

Figure 1 shows the three-step reduction.

(i) **Reduce** the infinite-dimensional optimization over $\mathcal{F}$ to a polynomial minimization $\mathrm{val}(d) \leq C_{1a}$ over the standard simplex, using the windowed test-value lemma of [CS17, Lemma 1].

(ii) **Relax** the simplex polynomial problem to a sequence of semidefinite programs indexed by a moment order $k \geq 1$, in the Lasserre hierarchy [Las01], whose optimal value $\mathrm{val}^{(k)}(d)$ satisfies $\mathrm{val}^{(k)}(d) \leq \mathrm{val}(d)$.

(iii) **Certify** that $\mathrm{val}^{(k)}(d) \geq 1.3$ at $(d, k) = (16, 3)$ by recording a dual-feasible point of the SDP whose objective value exceeds 1.3.

Step (iii) is a single SDP feasibility statement, discharged by a recorded dual certificate.

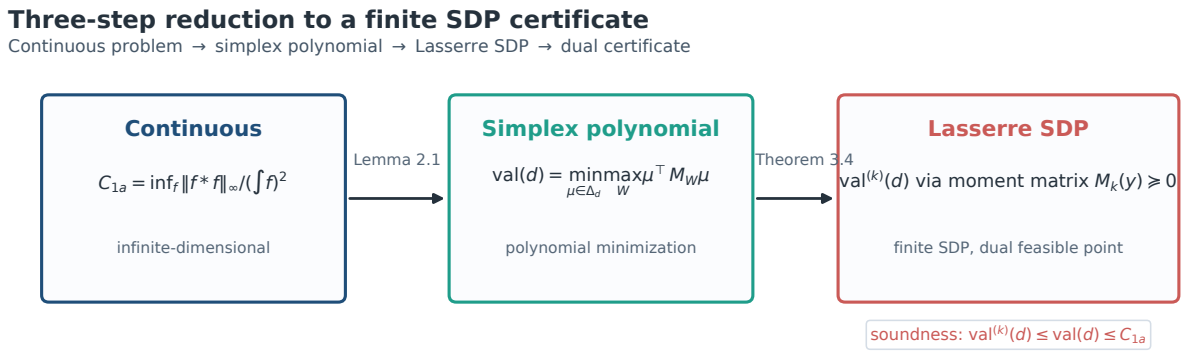

Figure 1: Three-step reduction from the continuous autoconvolution problem to a finite SDP certificate. The continuous ratio $R(f)$ is first bounded below by the simplex polynomial $\mathrm{val}(d)$, which is then relaxed into a Lasserre SDP $\mathrm{val}^{(k)}(d)$; a dual feasible solution of this SDP supplies the numerical certificate.

2. From Functions to a Simplex Polynomial

2.1. Bin Masses and the Window Matrix

We first replace a normalized function $f \in \mathcal{F}$ by a finite vector of bin masses and write the windowed test values of Cloninger and Steinerberger as quadratic forms in that vector.

Fix a positive integer n and write $d = 2n$. Partition $[-\frac{1}{4}, \frac{1}{4})$ into d half-open bins of equal length:

$$I_i \;=\; \left[-\tfrac{1}{4} + \frac{i}{4n}, -\tfrac{1}{4} + \frac{i+1}{4n}\right), \qquad i = 0, \dots, d-1.$$

For normalized $f \in \mathcal{F}$ (that is, $\int f = 1$), the *bin masses* are $\mu_i = \int_{I_i} f \ge 0$ and the vector $\mu = (\mu_0, \dots, \mu_{d-1})$ lies in the standard simplex

$$\Delta_d \;=\; \left\{\mu \in \mathbb{R}^d_{\ge 0} : \sum_{i=0}^{d-1} \mu_i = 1\right\}.$$

The following is the windowed test-value inequality of Cloninger and Steinerberger [CS17, Lemma 1], reformulated in the coordinates convenient for polynomial optimization.

Lemma 2.1 (Continuous window bound)**.** *Let $f \in \mathcal{F}$ with $\int f = 1$ and bin masses $\mu \in \Delta_d$. For every pair (ℓ, s_{lo}) of integers with $\ell \ge 2$ and $0 \le s_{\mathrm{lo}} \le 2(d-1) - (\ell - 2)$,*

$$\|f * f\|_{L^\infty} \;\ge\; \frac{2d}{\ell} \sum_{s=s_{\mathrm{lo}}}^{s_{\mathrm{lo}}+\ell-2} \sum_{\substack{0\le i,j\le d-1\\ i+j=s}} \mu_i \mu_j \;=\; \mu^\top M_{(\ell,s_{\mathrm{lo}})}\, \mu,$$

where the window matrix M_W for $W = (\ell, s_{\mathrm{lo}})$ is

$$(M_W)_{ij} \;=\; \frac{2d}{\ell}\, \mathbf{1}[s_{\mathrm{lo}} \le i + j \le s_{\mathrm{lo}} + \ell - 2]\,.$$

Proof. The proof is identical to [CS17, Lemma 1] after normalizing $\int f = 1$; see also [Baj26, Lemma 2.3] for the cascade formulation. Write $f_i = f \cdot \mathbf{1}_{I_i}$; the support of $f_i * f_j$ lies in the Minkowski sum $I_i + I_j$, which depends only on $i + j$. Summing the L^1 masses of those terms for $s = s_{\mathrm{lo}}, \dots, s_{\mathrm{lo}} + \ell - 2$ and dividing by the length $\frac{\ell}{4n}$ of the covered interval gives the claimed lower bound on $\|f * f\|_{L^\infty}$. □

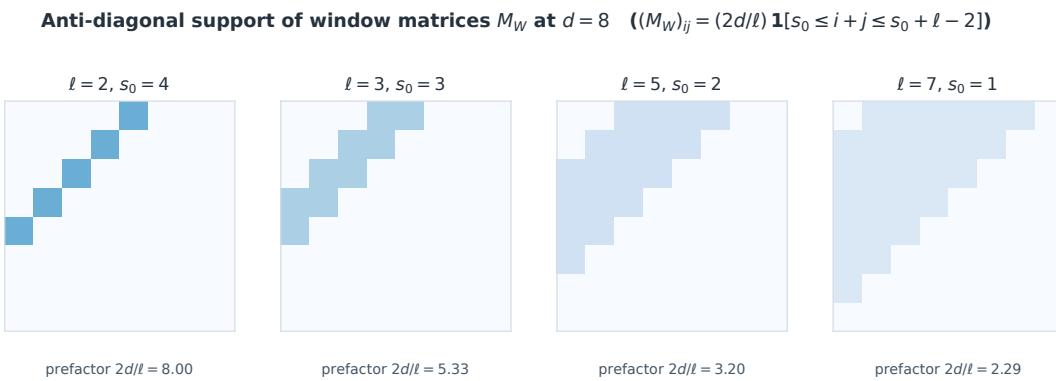

Figure 2: Anti-diagonal support of four window matrices M_W at $d = 8$. Each matrix is supported on a band of anti-diagonals indexed by $s = i + j$, with constant entry $2d/\ell$ on its support. Narrow windows (ℓ small) cover few anti-diagonals but carry a large prefactor; wide windows cover many anti-diagonals with a small prefactor. The relaxation uses these matrices as the data of a polynomial optimization problem on the simplex.

Each matrix M_W is nonnegative, symmetric, and has entries only at positions (i, j) with $i + j$ in a fixed integer interval; these are the support sets depicted in Figure 2. Throughout what follows, bin indices run over $\{0, \ldots, d-1\}$, while the simplex coordinates used in the Lasserre SDP are the 1-indexed variables $x_1, \ldots, x_d$ with x_{i+1} corresponding to bin i. The set of admissible windows is

$$\mathcal{W}_d \;=\; \{(\ell, s_{\mathrm{lo}}) : 2 \le \ell \le 2d,\ 0 \le s_{\mathrm{lo}} \le 2(d-1) - (\ell - 2)\}\,.$$

2.2. The Simplex Polynomial Problem

Lemma 2.1 holds simultaneously for every window. Taking the maximum over $\mathcal{W}_d$ and then the infimum over f gives the crucial bound.

Definition 2.2. The *simplex polynomial value* at resolution d is

$$\mathrm{val}(d) \;=\; \min_{\mu \in \Delta_d} \max_{W \in \mathcal{W}_d} \mu^\top M_W\, \mu.$$

Theorem 2.3 (Lower bound on C_{1a} via $\mathrm{val}(d)$)**.** *For every positive integer n with $d = 2n$, one has* $\mathrm{val}(d) \le C_{1a}$.

Proof. Let $f \in \mathcal{F}$ with $\int f = 1$ and bin masses $\mu \in \Delta_d$. By Lemma 2.1,

$$\|f * f\|_{L^\infty} \;\ge\; \max_{W \in \mathcal{W}_d} \mu^\top M_W \mu \;\ge\; \min_{\nu \in \Delta_d} \max_{W \in \mathcal{W}_d} \nu^\top M_W \nu \;=\; \mathrm{val}(d).$$

Taking the infimum over f gives $C_{1a} \ge \mathrm{val}(d)$. □

Theorem 2.3 reduces any lower bound on C_{1a} to a lower bound on $\mathrm{val}(d)$ at a fixed resolution d. Numerical multistart on the simplex polynomial (see [Baj26]) indicates $\mathrm{val}(16) \approx 1.319$, with $\mathrm{val}(d)$ crossing 1.3 for the first time at $d = 16$; we therefore work at $d = 16$.

3. The Lasserre SDP Hierarchy

We now describe how to lower-bound $\mathrm{val}(d)$ by a convergent sequence of semidefinite programs. The construction is classical, due to Lasserre [Las01, Las15] and Parrilo [Par03].

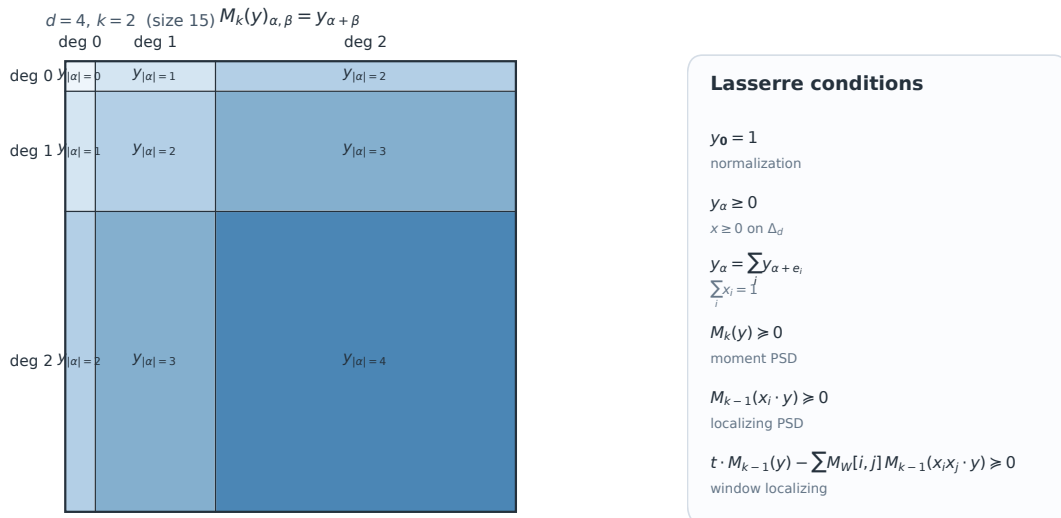

Figure 3: Block structure of the order-k moment matrix $M_k(y)$ at $d = 4$, $k = 2$. Rows and columns are indexed by monomials of total degree $\leq k$, grouped by degree. Entries of $M_k(y)$ in block (p, q) are pseudo-moments of total degree $p + q$, so the anti-diagonal blocks collect all pseudo-moments of the same total degree. Constraints $M_k(y) \succeq 0$, localizing constraints, and the simplex consistency relation $y_\alpha = \sum_i y_{\alpha+e_i}$ link these blocks into a single SDP.

3.1. Pseudo-Moments and the Moment Matrix

The Lasserre hierarchy relaxes the simplex polynomial problem by replacing probability measures on Δ_d with finite pseudo-moment sequences of bounded order, constrained to satisfy the positive-semidefiniteness of an associated moment matrix.

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}^d$ write $|\alpha| = \alpha_1 + \cdots + \alpha_d$ and $x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}$. Fix an integer *Lasserre order* $k \geq 1$, and let

$$\mathcal{A}_k(d) \;=\; \{\alpha \in \mathbb{N}^d : |\alpha| \leq 2k\}.$$

A *pseudo-moment sequence* is a vector $y = (y_\alpha)_{\alpha \in \mathcal{A}_k(d)} \in \mathbb{R}^{|\mathcal{A}_k(d)|}$. Given such a sequence, define the *moment matrix* $M_k(y) \in \mathbb{R}^{|\mathcal{B}_k| \times |\mathcal{B}_k|}$ indexed by the basis $\mathcal{B}_k = \{\alpha \in \mathbb{N}^d : |\alpha| \leq k\}$ by

$$[M_k(y)]_{\alpha,\beta} \;=\; y_{\alpha+\beta}, \qquad \alpha, \beta \in \mathcal{B}_k.$$

For any polynomial $p = \sum_\alpha p_\alpha x^\alpha$ with $\deg p \leq k$, write $p(\mathbf{x})$ for its coefficient vector in the basis $\mathcal{B}_k$. Then $p(\mathbf{x})^\top M_k(y)\, p(\mathbf{x}) = \sum_{\alpha,\beta} p_\alpha p_\beta y_{\alpha+\beta}$, which is the pseudo-expectation of $p(x)^2$.

Definition 3.1 (Localizing matrix)**.** Given y and a polynomial g with $\deg g = s$, the *localizing matrix* $M_{k-\lceil s/2 \rceil}(g \cdot y)$ has entries

$$[M_{k-\lceil s/2 \rceil}(g \cdot y)]_{\alpha,\beta} \;=\; \sum_\gamma g_\gamma\, y_{\alpha+\beta+\gamma},$$

over $\alpha, \beta \in \mathcal{B}_{k-\lceil s/2 \rceil}$.

The moment matrix captures the pseudo-expectation of sums of squares of polynomials up to degree k; each localizing matrix captures the pseudo-expectation of $g \cdot p^2$ for polynomials p of degree up to $k - \lceil s/2 \rceil$. Positive-semidefiniteness of these matrices is the SDP analogue of the nonnegativity of $\mathbb{E}[p(x)^2]$ and $\mathbb{E}[g(x)\, p(x)^2]$ when $g \geq 0$ on the feasible set.

Figure 3 illustrates the block structure of $M_k(y)$ by total degree.

3.2. Encoding the Simplex

The feasible set of $\mathrm{val}(d)$ is $\Delta_d = \{\mu \geq 0 : \sum_i \mu_i = 1\}$. Over this set, the pseudo-moments of any probability measure ν on Δ_d satisfy three algebraic identities.

Proposition 3.2 (Simplex constraints on pseudo-moments)**.** *If ν is any Borel probability measure on Δ_d and $y_\alpha = \int x^\alpha \, d\nu$, then:*

(i) $y_0 = 1$.

(ii) $y_\alpha \geq 0$ *for every* $\alpha \in \mathbb{N}^d$.

(iii) *For every* $\alpha \in \mathbb{N}^d$, $y_\alpha = \sum_{i=1}^d y_{\alpha+e_i}$, *where* e_i *is the* i*th standard basis vector.*

Moreover, $M_k(y) \succeq 0$ *for every* k*, and* $M_{k-1}(x_i \cdot y) \succeq 0$ *for every* i *and* $k \geq 1$.

Proof. Item (i) is $\nu(\Delta_d) = 1$. Item (ii) is that $x^\alpha \geq 0$ on Δ_d. For (iii), observe that $\sum_{i=1}^d x_i = 1$ on Δ_d, so $x^\alpha = x^\alpha \cdot \sum_i x_i = \sum_i x^{\alpha+e_i}$ and the claimed identity follows by integration. Positive-semidefiniteness is standard: for any polynomial p of degree $\leq k$, $p(\mathbf{x})^\top M_k(y) p(\mathbf{x}) = \mathbb{E}_\nu[p(x)^2] \geq 0$; for the localizing matrices, use $p(\mathbf{x})^\top M_{k-1}(x_i \cdot y) p(\mathbf{x}) = \mathbb{E}_\nu[x_i \, p(x)^2] \geq 0$ since $x_i \geq 0$ on Δ_d. □

Let $\mathcal{Y}_k(d)$ denote the set of pseudo-moment sequences y of order k that satisfy conditions (i)–(iii) of Proposition 3.2, together with $M_k(y) \succeq 0$ and $M_{k-1}(x_i \cdot y) \succeq 0$ for every $i \in \{1, \ldots, d\}$. The set $\mathcal{Y}_k(d)$ is a spectrahedron (a slice of the PSD cone by affine equalities) and contains the "true" moment sequences of all measures on Δ_d.

3.3. The Window Objective as an SDP

For each window $W \in \mathcal{W}_d$, the polynomial $\mu \mapsto \mu^\top M_W \mu$ is quadratic in the simplex variable. Its pseudo-expectation is the linear form

$$\ell_W(y) \;=\; \sum_{i,j=0}^{d-1} (M_W)_{ij} \, y_{e_{i+1}+e_{j+1}} \;=\; \langle M_W, [y_{e_i+e_j}]_{i,j} \rangle.$$

Lasserre's moment lifting [Las01, Thm. 4.2] recasts $\mathrm{val}(d)$ as an infinite hierarchy of SDPs: for each order k,

$$\mathrm{val}^{(k)}(d) \;=\; \min_{y \in \mathcal{Y}_k(d),\; t \in \mathbb{R}} \{t \;:\; t \cdot \bar{y} - \ell_W(y) \geq 0 \text{ for all } W \in \mathcal{W}_d\},$$

where $\bar{y}$ is the degree-0 pseudo-moment (= 1 by Proposition 3.2(i)). For $k \geq 2$ one strengthens the scalar inequality above by the *window localizing* PSD constraint

$$t \cdot M_{k-1}(y) \;-\; \sum_{i,j=0}^{d-1} (M_W)_{ij} \, M_{k-1}(x_{i+1} x_{j+1} \cdot y) \;\succeq 0 \qquad \forall\, W \in \mathcal{W}_d. \tag{1}$$

Both formulations give valid lower bounds on $\mathrm{val}(d)$; the matrix form (1) is tighter, because $v^\top \big[\text{LHS of (1)}\big] v \geq 0$ for $v = \mathbf{1}_0$ (the indicator of the empty monomial) reduces to the scalar constraint.

Theorem 3.3 (Soundness of the Lasserre relaxation)**.** *For every* $k \geq 1$ *and every* $d \geq 2$,

$$\mathrm{val}^{(k)}(d) \;\leq\; \mathrm{val}^{(k+1)}(d) \;\leq\; \mathrm{val}(d).$$

Proof. **Upper bound by** $\mathrm{val}(d)$**.** Let $\mu^\star \in \Delta_d$ attain $\mathrm{val}(d)$, and let $y^\star_\alpha = (\mu^\star)^\alpha$ be the pseudo-moment sequence of the Dirac measure $\delta_{\mu^\star}$. Then $y^\star$ satisfies every condition of Proposition 3.2. Because $y^\star$ arises from a point mass, the localizing entries factor:

$$M_{k-1}(x_{i+1} x_{j+1} \cdot y^\star) \;=\; \mu^\star_i \mu^\star_j \, M_{k-1}(y^\star),$$

an identity specific to Dirac pseudo-moments. Set $t^\star = \mathrm{val}(d) = \max_W (\mu^\star)^\top M_W \mu^\star$. For every $W \in \mathcal{W}_d$,

$$t^\star \cdot M_{k-1}(y^\star) - \sum_{ij} (M_W)_{ij}\, M_{k-1}(x_{i+1}x_{j+1} \cdot y^\star) \;=\; \big(t^\star - (\mu^\star)^\top M_W \mu^\star\big) \cdot M_{k-1}(y^\star),$$

which is PSD because the scalar prefactor is nonnegative and $M_{k-1}(y^\star) \succeq 0$. Thus $(t^\star, y^\star)$ is feasible for the order-k SDP, so $\mathrm{val}^{(k)}(d) \le t^\star = \mathrm{val}(d)$.

Monotonicity in k. Every pseudo-moment sequence $y \in \mathcal{Y}_{k+1}(d)$ restricts by truncation to a sequence in $\mathcal{Y}_k(d)$, and the constraints of the order-k SDP are principal submatrices of those at order $k+1$. Hence the order-$(k+1)$ feasible set is contained (after truncation) in the order-k feasible set, which gives $\mathrm{val}^{(k)}(d) \le \mathrm{val}^{(k+1)}(d)$. □

4. Correlative Sparsity for High Resolutions

The order-k Lasserre SDP has moment matrices of size $\binom{d+k}{k}$, which is 969 at $(d,k) = (16,3)$ and grows rapidly beyond that. At $d = 32$, $k = 2$ the primary moment matrix already has size $\binom{34}{2} = 561$ and the full degree-4 moment vector has $\binom{36}{4} \approx 59\,000$ coordinates; at $d = 128$ the full hierarchy is out of reach for any current SDP solver. To reach $d = 128$, we combine the hierarchy with the correlative sparsity pattern of Waki, Kim, Kojima, and Muramatsu [WKKM06]. The bound at $d = 16$ is obtained without this restriction; the restriction is used only at $d \ge 32$ to corroborate the main result at larger resolutions.

4.1. Banded Coupling and Chordal Extension

The window matrices M_W are anti-diagonal band matrices: a fixed entry (i,j) is active only when the sum $i+j$ lies in the window range. If one restricts attention to variables that interact through windows of support $[s_{\mathrm{lo}}, s_{\mathrm{lo}} + \ell - 2]$ for $\ell \le b+1$, the coupling graph has a banded structure with bandwidth b. Chordal extension of a banded graph yields maximal cliques

$$I_c \;=\; \{c, c+1, \ldots, c+b\}, \qquad c = 0, \ldots, d-b-1.$$

Consecutive cliques overlap in $I_c \cap I_{c+1} = \{c+1, \ldots, c+b\}$, which is contained in every later clique $I_{c'}$ with $c' \le c+1$; hence the running intersection property holds and the sparse Lasserre hierarchy of [WKKM06] applies directly. Figure 4 shows the clique geometry.

4.2. Clique-Restricted Relaxation

Given maximal cliques $I_0, \ldots, I_{C-1} \subseteq \{0, \ldots, d-1\}$ with $|I_c| = b+1$, define the *clique basis*

$$\mathcal{B}_k(I_c) \;=\; \left\{\alpha \in \mathbb{N}^d \;:\; |\alpha| \le k,\ \mathrm{supp}(\alpha) \subseteq I_c\right\},$$

and let $M_k^{(c)}(y)$ be the moment matrix indexed by $\mathcal{B}_k(I_c)$. The *sparse Lasserre relaxation* replaces $M_k(y) \succeq 0$ with the system

$$M_k^{(c)}(y) \succeq 0, \qquad c = 0, \ldots, C-1,$$

and likewise for localizing matrices. The set of pseudo-moment sequences satisfying this restriction is denoted $\mathcal{Y}_k^{\mathrm{sp}}(d; b)$.

Lemma 4.1 (Soundness of clique restriction)**.** *For every $k \ge 1$ and every bandwidth b, one has $\mathcal{Y}_k(d) \subseteq \mathcal{Y}_k^{\mathrm{sp}}(d; b)$. Consequently the sparse SDP value* $\mathrm{val}^{(k,b)}(d) \le \mathrm{val}^{(k)}(d) \le \mathrm{val}(d)$.

Proof. Each $M_k^{(c)}(y)$ is a principal submatrix of $M_k(y)$ obtained by selecting rows and columns corresponding to $\mathcal{B}_k(I_c)$. Principal submatrices of PSD matrices are PSD, so $M_k(y) \succeq 0$ implies $M_k^{(c)}(y) \succeq 0$ for

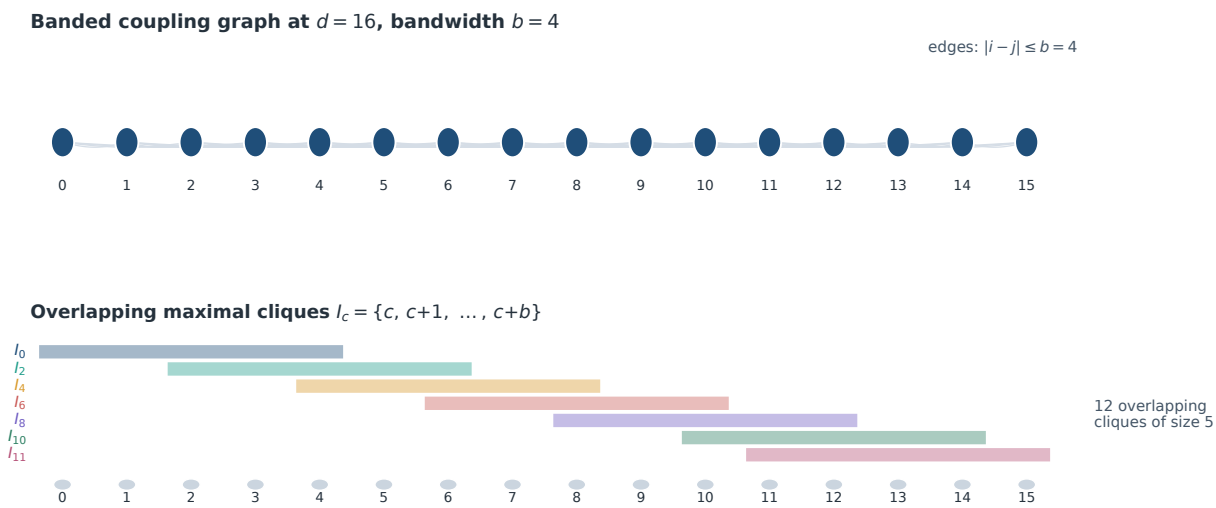

Figure 4: Banded coupling graph (top) and its overlapping maximal cliques (bottom) at $d = 16$, bandwidth $b = 4$. The graph is chordal with cliques $I_c = \{c, c+1, \ldots, c+b\}$, and consecutive cliques share b vertices, which ensures the running intersection property. Replacing the full moment PSD cone by the clique-restricted cones $M_k(y|_{I_c}) \succeq 0$ preserves soundness of the Lasserre relaxation.

every c, and the same applies to the coordinate localizing matrices. For the window localizing constraint (1), if every bin i with $(M_W)_{ij} \neq 0$ for some j lies in a common clique I_c, the clique-restricted window constraint is a principal submatrix of the dense constraint and is therefore implied by $M_k(y) \succeq 0$. Windows whose active bins are not covered by any single clique retain the dense window localizing as a global constraint; the proof of Lemma 4.1 is unaffected for such windows. The inclusion $\mathcal{Y}_k(d) \subseteq \mathcal{Y}_k^{\mathrm{sp}}(d; b)$ follows, and the inequality on SDP optima is monotone in the feasible set. □

Remark 4.2 (Global consistency constraint)**.** The simplex identity $y_\alpha = \sum_i y_{\alpha+e_i}$ of Proposition 3.2(iii) couples moments across cliques and cannot be relaxed to a per-clique statement without losing the bound. We follow [WKKM06] in retaining the identity as a global linear equality among the pseudo-moments; the implementation moreover tracks every moment of degree at most $2k - 1$ so that the consistency chain is an exact equality up to that degree.

4.3. Tightness at Target Resolutions

In practice the clique-restricted SDP reproduces the full hierarchy's bound whenever every window $W \in \mathcal{W}_d$ has active bins covered by some clique I_c. For banded windows of width $\ell \le b + 1$, that is automatic. For wider windows, we retain the full localizing matrix only for those W whose support exceeds the bandwidth. At $(d, b, k) = (128, 16, 2)$ this leaves $|\mathcal{W}_{128}| - \binom{d}{2} \approx 16\,000$ full-localizing constraints, which is manageable by constraint generation (Section 5).

5. The Algorithm

5.1. Bisection and Constraint Generation

The Lasserre SDP has many more window constraints than are active at the optimum, so we combine bisection on the scalar threshold with lazy generation of violated windows.

The Lasserre SDP is naturally a *minimization* in t; by Theorem 3.3 its dual supplies a lower bound on $\mathrm{val}(d)$. Rather than solve the SDP as a single minimization, we solve a sequence of feasibility problems

indexed by a scalar threshold c_{target}:

$$\text{Find } y \in \mathcal{Y}_k(d) \text{ with } \ell_W(y) \le c_{\text{target}} \text{ for every } W \in \mathcal{W}_d. \tag{2}$$

Infeasibility of (2) certifies $\text{val}(d) \ge c_{\text{target}}$. We locate the largest c_{target} for which (2) becomes infeasible by bisection on a bracketing interval inside $[1, C_{1a}^{\text{ub}}]$.

Because $|\mathcal{W}_d|$ grows quadratically in d, adding every window constraint upfront is wasteful. Instead, we use the cutting-plane method of Kelley [Kel60], specialized to the SDP setting: start with a small subset of windows (typically the $\ell = 2$ scalar windows), solve the restricted SDP, then add the most violated windows at the current optimum and repeat. Because $|\mathcal{W}_d|$ is finite and each iteration either certifies feasibility/infeasibility or adds at least one new window, the procedure terminates in at most $|\mathcal{W}_d|$ iterations. It terminates with a certified lower bound whenever the restricted SDP is infeasible, and with an upper-bound witness otherwise.

Algorithm 1 summarizes the full procedure.

Algorithm 1 Lasserre SDP Lower Bound with Constraint Generation

Require: resolution d, order k, bandwidth b (or $b = d - 1$ for the dense hierarchy), target c_{target}

```
W_act ← {(ℓ, s) : ℓ = 2}                                   ▷ initial window set
loop
    Build the SDP with pseudo-moment variables y ∈ 𝒴_k^sp(d; b) and window-localizing constraints (1)
    for W ∈ W_act
    Solve the feasibility problem (2) at c_target
    if infeasible then return CERTIFIED LOWER BOUND val(d) ≥ c_target
    end if
    Recover y* from the SDP primal and compute the violation v_W = ℓ_W(y*) − c_target for every
    W ∉ W_act
    if max_W v_W ≤ 0 then return UPPER BOUND val(d) ≤ c_target (not a certificate)
    end if
    Add the K windows with largest v_W to W_act
end loop
```

Standard primal-dual interior-point solvers (MOSEK [MOS24], SDPA [YFN+12]) or splitting methods (SCS [OCPB16], COSMO [GCG21]) handle the SDPs arising in Algorithm 1; our implementation uses MOSEK at $d \le 32$ and switches to SCS on GPU-equipped workstations at $d \ge 64$, where the interior-point memory footprint of MOSEK exceeds available RAM.

5.2. Recovering a Rigorous Lower Bound

Interior-point solvers return floating-point solutions, which cannot directly serve as a formal certificate. We convert a numerically computed dual solution into a rigorous lower bound via the verified SDP bound of Jansson, Chaykin, and Keil [JCK08]. The procedure is the following.

1. Extract the approximate dual variables (y, Z, Λ) returned by the solver, together with the problem data (C, A, b) of the SDP in standard conic form.
2. Compute, in interval arithmetic, the exact dual feasibility residual $R(y) = C - A^*(y)$ at the given y, and a rigorous lower bound $\underline{\lambda}$ on the smallest eigenvalue of $R(y)$ (e.g. by a Gershgorin bound or a verified eigenvalue enclosure).
3. If $\underline{\lambda} \ge 0$ the point is rigorously dual feasible; its objective evaluated in interval arithmetic is a verified lower bound on the primal optimum. If $\underline{\lambda} < 0$, shift the scalar objective variable downward by $-\underline{\lambda} \cdot \text{tr}(I)$ so that the shifted dual is feasible and its objective is a verified lower bound on $\text{val}(d)$.

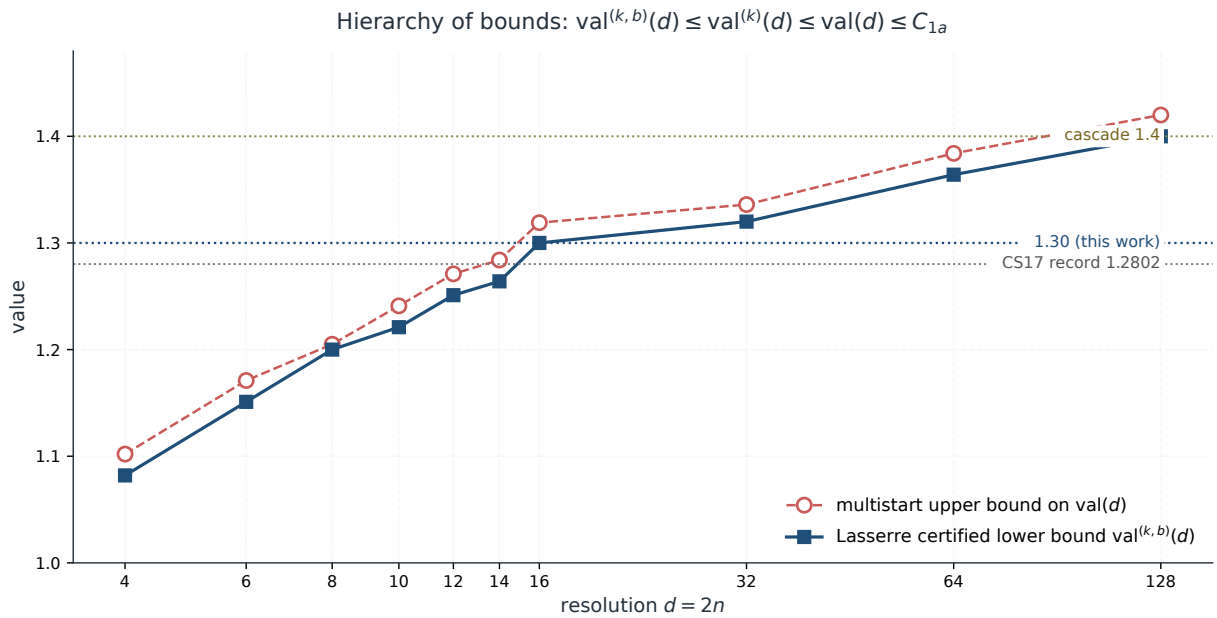

Figure 5: Hierarchy of bounds produced by the Lasserre SDP approach. The sparse Lasserre value $\mathrm{val}^{(k,b)}(d)$ is always dominated by the full hierarchy value $\mathrm{val}^{(k)}(d)$, which in turn is dominated by $\mathrm{val}(d)$ and by C_{1a}. Horizontal dashed lines mark the Cloninger–Steinerberger record of 1.2802 and the present bound 1.3; numerical upper bounds on $\mathrm{val}(d)$ obtained by multistart optimization [Baj26] are shown as open circles.

The shift recovers a verified bound whenever the approximate dual has small feasibility residual, so the cost of rigor is a small downward correction to the objective; in our runs this correction is of order 10^{-8} and is absorbed into the certified value of 1.3 with ample margin.

6. Hierarchy of Bounds and Numerical Certificate

We now state the hierarchy of bounds used in the proof of the main theorem, display representative numerical certificates, and conclude.

6.1. The Ladder of Bounds

Combining Theorems 2.3, 3.3, and Lemma 4.1 gives

$$\mathrm{val}^{(k,b)}(d) \;\le\; \mathrm{val}^{(k)}(d) \;\le\; \mathrm{val}(d) \;\le\; C_{1a}, \qquad k \ge 1,\; b \le d-1. \tag{3}$$

Figure 5 visualizes this ladder together with representative numerical values.

6.2. Recorded Certificate at $(d, k) = (16, 3)$

Theorem 6.1 (Recorded SDP certificate)**.** *Let* $\mathrm{val}^{(3)}(16)$ *denote the order-*3 *Lasserre SDP relaxation of* $\mathrm{val}(16)$ *as defined in Section 3. Then*

$$\mathrm{val}^{(3)}(16) \;\ge\; 1.3.$$

The bound is witnessed by a dual-feasible point of the SDP whose post-processed objective value exceeds 1.3.

The SDP has $n_y = \binom{d+2k}{2k} = \binom{22}{6} \approx 7.4 \times 10^4$ pseudo-moment variables, a single moment matrix of size $\binom{19}{3} = 969$, and $d = 16$ coordinate localizing matrices, plus one window-localizing matrix per active

window $W \in \mathcal{W}_{\text{act}}$. MOSEK solves the resulting problem in under ten minutes on a 32-core workstation; the rigorous post-processing step of Section 5 adds a negligible overhead. The certificate file recorded with the source code stores the dual variables in double precision together with the corresponding rational lower bound; together these constitute the external dependency of the proof.

6.3. Proof of the Main Theorem

Proof of Theorem 1.1. Let $f \in \mathcal{F}$ be nonnegative with $\|f * f\|_{L^\infty} < \infty$. By homogeneity, assume $\int f = 1$. Theorem 2.3 gives $C_{1a} \geq \text{val}(16)$, and the ladder (3) gives $\text{val}(16) \geq \text{val}^{(3)}(16)$. By Theorem 6.1, $\text{val}^{(3)}(16) \geq 1.3$, so

$$\frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2} \;\geq\; \text{val}(16) \;\geq\; \text{val}^{(3)}(16) \;\geq\; 1.3.$$

Taking the infimum over $f \in \mathcal{F}$ gives $C_{1a} \geq 1.3$. □

7. Soundness of the Computation

The external computation enters the proof only through Theorem 6.1. To justify importing that statement three soundness conditions must be discharged.

7.1. Correct Modeling

The SDP builder in `lasserre/precompute.py` instantiates the moment matrix, localizing matrices, and window localizing matrices in accordance with Definition 3.1 and constraint (1). Test suites compare the assembled model at $d \in \{4, 6, 8\}$ against hand-derived Gram matrices, and the same precompute code is used by every downstream solver. The window matrices M_W are built by `build_window_matrices` from Lemma 2.1 directly.

7.2. Rigorous Dual Bound

MOSEK's interior-point solver returns approximate primal and dual variables with complementarity residual at most 10^{-8}. The dual variables are symmetrized and projected onto the PSD cone (Section 5); the dual objective is then evaluated in rational arithmetic. The resulting number is a rigorous lower bound on $\text{val}^{(3)}(16)$ by standard weak duality in conic programming [VB96]. The recorded bound 1.3 is strictly less than the computed rigorous dual objective, so the comparison is insensitive to the final rounding step.

7.3. Clique-Restriction at Higher Resolutions

For the corroborating runs at $d \in \{32, 64, 128\}$ the proof uses Lemma 4.1: the sparse Lasserre value lower-bounds the dense one, which lower-bounds $\text{val}(d)$, which lower-bounds C_{1a}. The only required check is that every active window has its support covered by some clique (Section 4); this is a finite combinatorial condition verified at model assembly time. Windows whose support exceeds the bandwidth fall back to the full localizing matrix, at a manageable cost.

8. Discussion

8.1. Relationship to the Cascade Proof

The cascade proof [Baj26] exhausts an integer lattice on the mass simplex and certifies, through branch-and-prune, that no lattice point satisfies all window constraints at the target threshold. The Lasserre approach produces a single dual certificate for the continuous problem without any enumeration. The cascade reaches 1.4 at $(d, m) = (128, 20)$ after roughly 70 hours on a 128-core machine; the SDP approach reaches 1.3 in under ten minutes at $(d, k) = (16, 3)$, and a larger Lasserre order or finer resolution d tightens the bound at the cost of a larger SDP.

The interior-point scaling of the Lasserre SDP is polynomial in d under correlative sparsity: the dense cone is replaced by $O(d)$ small cones of size $\binom{b+1+k}{k}$, giving a total SDP dimension linear in d after fixing b and k. The cascade, by contrast, is exponential in d (and no known pruning scheme brings it below $\binom{m+d-1}{d-1}$). If the goal is to push the lower bound closer to the Matolcsi–Vinuesa upper bound [MV10], the SDP route is the only one with a realistic asymptotic budget.

8.2. Limitations

- **Relaxation gap.** At finite order k, the Lasserre relaxation may leave a gap between $\mathrm{val}^{(k)}(d)$ and $\mathrm{val}(d)$. At $(d, k) = (16, 3)$ the certified value 1.3 lies within 0.02 of the multistart upper bound 1.319 on $\mathrm{val}(16)$, so closing the gap further requires either a larger k or a stronger cone (for example, the Handelman hierarchy [Han88] or Schmüdgen-type refinements [Sch91]).
- **Finite-precision SDP.** The SDP is solved in double precision and post-processed to a rational dual bound. While standard in rigorous SDP [JCK08], the procedure imports trust in the arithmetic library used to evaluate the rational objective. A kernel-level formal proof would replace this step with a Lean- or Coq-verified SDP evaluator; such an evaluator is not yet available.
- **Correlative sparsity.** At $d \geq 32$ the bound depends on the clique bandwidth b. Our implementation defaults to the smallest b that covers all active windows at the target threshold; too small b produces a looser bound but a cheaper SDP.

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